



Selfie Dental AI Assistant

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Stakeholder: Zhuo Wang (Dental Assistant)

NEED STATEMENT

There is a need for an easy-to-use, affordable, accessible educational platform that helps users understand and track their oral health, encouraging early intervention and regular preventive care, particularly for populations with limited access to dental services.

Design Parameters

Design Input

1

Overall, the design must be accessible, affordable, and user-friendly while being flexible enough to integrate future diagnostic technologies.

2

Constraints

- System relies only on user input and smartphone photos;
- No X-rays
- Prototype uses open-source images due to limited access to proprietary datasets
- Platform must remain non-diagnostic
- Must comply with PIPEDA and App Store / Google Play policies

Assumptions

3

- Users can capture three basic intraoral photos using their smartphone.
- Users can follow simple instructions for photos and questionnaire responses.
 - Smartphone cameras provide adequate quality for visual screening.

Assumptions

1. The user accept the “Terms and Agreements” created, giving the application permission to collect necessary data.
2. The user will input clear intra/extra-oral images.
3. The user will answer the questionnaire honestly.
4. **The CNN has been trained on a diverse and representative dataset that captures real-world variability.**
5. Assuming Render provide secure TLS

Limitations

1. The dataset uses partially labeled data, meaning not all conditions are annotated for every image. Will limit how well the model learns relationships between multiple conditions.
2. The rule-based questionnaire simplifies complex pathology, which may limit diagnostic precision in borderline cases. .

Constraints

Time & Server Costs

1. Ethical and regulatory constraints

- Careful handling of identifiable images -> local processing & minimizing stored data

2. Technical constraints

- limited availability of labelled datasets
- need for lightweight models suitable for mobile deployment

3. Computational constraints

- Select compact backbones (MobileNetV3 and ResNet-18) to allow on-device inference while maintaining acceptable performance.

Backend Overview



Image



```
answers = {  
  "age_group": "35_to_55",  
  "dental_visits": ">1yr",  
  "hygiene_habits": "once_daily",  
  "diet_sugar": "frequently",  
  "staining_habits": "diet_only",  
  "dry_mouth": "yes",  
  "systemic_health": "no",  
  "trauma_history": "no",  
  "bleeding_gums": "sometimes",  
  "pain_sensitivity": "mild"  
}
```

Questionnaire
Answers



```
DIAGNOSIS REPORT  
=====
```

Detected: Calculus, Caries, Gingivitis	

Raw Probabilities:	
[!] Calculus	: 99.1%
[!] Caries	: 79.3%
[!] Gingivitis	: 99.5%
Tooth discoloration : 48.5%	
=====	

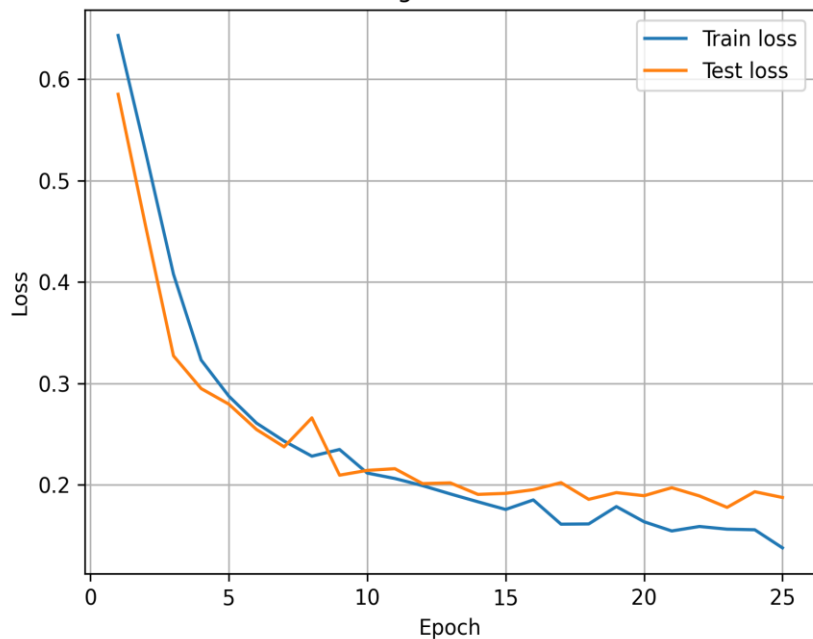
Results

Visual Model Details

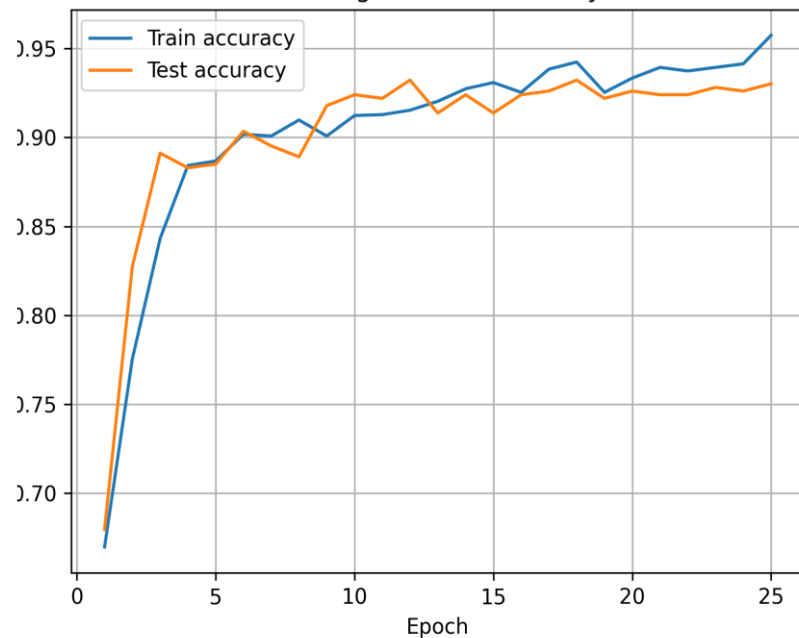
- **MobileNetV3 Small** network with transfer learning
- **Image Preprocessing**
 - resized to 224 by 224 pixels
 - converted to tensors
 - normalized using ImageNet statistics
 - Training Data augmented with random rotation, horizontal flipping, and brightness adjustments
- **Masked multi-label approach**
 - Healthy [0, 0, 0, 0]
 - Disease A [1, -1, -1, -1]
- **Model Architecture**
 - A frozen backbone and a custom classifier head
 - Trained with masked binary cross-entropy loss and Adam optimizer.

Visual Model Results

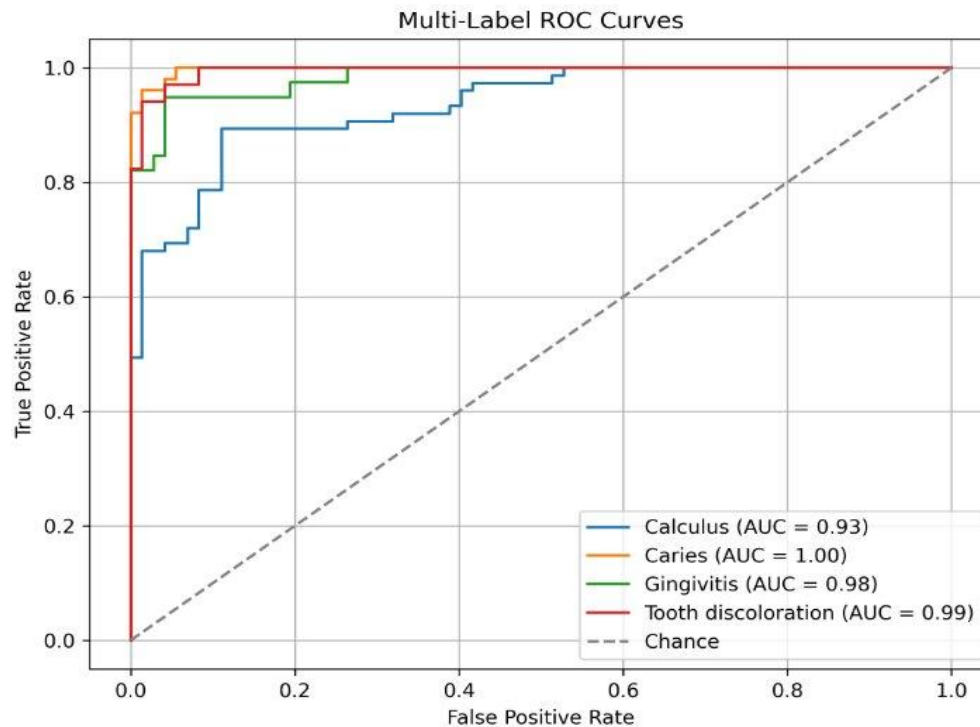
Training and Test Loss



Training and Test Accuracy



Visual Model Results



Design Iterations

- **POC** – Binary classification model to detect Dental Calculus
- **Expansion**
 - Multi-label classification (Calculus, Caries, Gingivitis, Tooth Discoloration)
 - ~400 manually labeled images [1,1,0,0]
- **Challenges**
 - Label Mismatch caused low accuracy
 - Solution: Instead of complete overhaul used Masked Labels (-1)
 - Benefit: Reuse existing images and improved scalability
 - Data Imbalance
 - Solution: Data augmentation, deliberate data collection for underrepresented populations
 - Ambiguous visual features
 - Fusion weight tuning and threshold tuning
 - Solution: Extensive research, stakeholder feedback and testing with sample images
- **Optimization**
 - Experimented with different architectures, including ResNet18 and MobileNetV3
 - Experimented with different classification layer architecture
 - Experimented with model training, number of epochs, different optimizers etc.

Pre-Processing Validation

Input Image



Quality Check (resolution, brightness, blur, contrast,)



Pre-filter Model (Detect teeth + gumline)



Main Model (Disease Prediction)

- Prevents invalid inputs
- Ensures images contain teeth & gums
- Improves reliability of predictions
- Reduces false positives



Testing

- Tested on **unseen images** (20+, covering a variety of condition combinations with different severity)
- Labels provided by **stakeholder**
- Strong generalization performance
- Performance improves with **questionnaire fusion**
- High sensitivity → detects most disease cases



Healthy [0, 0, 0, 0]

```
"detected_conditions": [  
  "Gingivitis"  
],  
"image_probabilities": {  
  "Calculus": 0.24341724812984467,  
  "Caries": 0.13035152852535248,  
  "Gingivitis": 0.513963520526886,  
  "Tooth Discoloration": 0.011649573221802711  
},
```

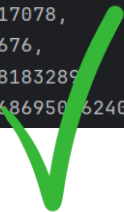
Testing Continued

[Calculus, Caries, Gingivitis, Tooth Discoloration]



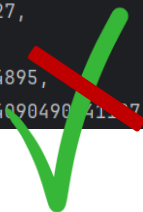
[1, 1, 1, 1]

```
"image_probabilities": {  
  "Calculus": 0.9614011645317078,  
  "Caries": 0.6847766041755676,  
  "Gingivitis": 0.9706622958183289,  
  "Tooth Discoloration": 0.686950624076843
```



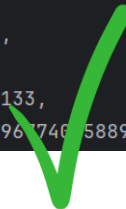
[1, 0, 1, 1]

```
"image_probabilities": {  
  "Calculus": 0.9432870149612427,  
  "Caries": 0.6151801943778992,  
  "Gingivitis": 0.9028361439704895,  
  "Tooth Discoloration": 0.756409049041127
```



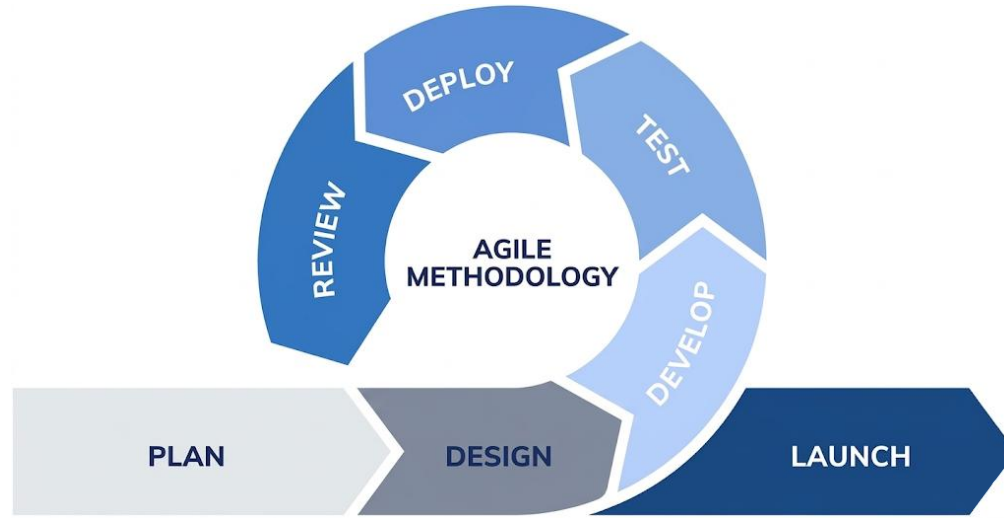
[0, 0, 0, 0]

```
"image_probabilities": {  
  "Calculus": 0.3173515796661377,  
  "Caries": 0.02322389744222164,  
  "Gingivitis": 0.13639000058174133,  
  "Tooth Discoloration": 0.0480196174658899
```



LIVE DEMO TIME

Anyone would like to volunteer for this?



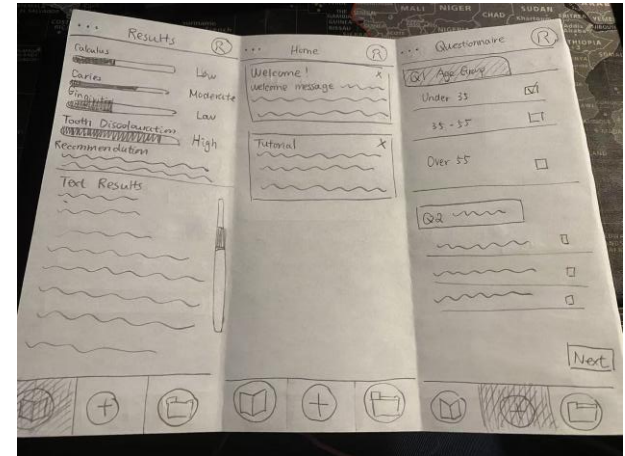
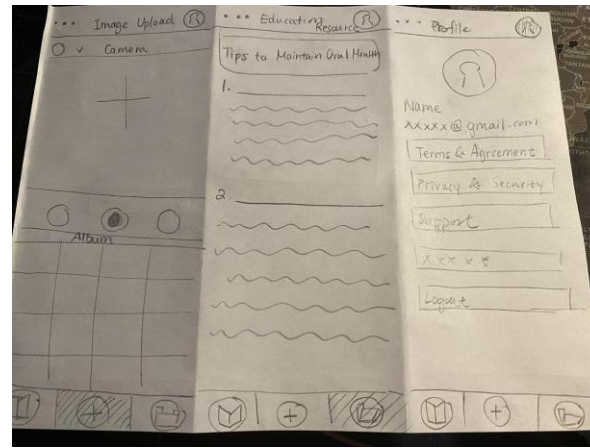
Wireframe Prototype-> Interactive Prototype -> Front End Framework & API

FRONT END DEVELOPMENT CYCLE

1.0 Wireframe Prototype

Paper Prototype

- Conceptual Model
- Functionality
- Navigation and Task Flow
- Terminology
- Screen contents



Very useful for in person group interview
Confusing for Online Surveys

https://docs.google.com/forms/d/1MgCfZQbZHiAKRVsUy_1-Z9DcKyBplt7ecBf_fMBUU4w/edit#responses

2.0 Interactive Prototype Review

The **online survey** was not really helpful

- Looks too overwhelming
- Confusing without guidance
- Complaints: poor drawing, looks ugly...

In Person Review:

- Want process to be linear and simple
- Too much text, crowded information
- Good general App layout, need to work on details

Any other suggestions

2 responses

I will more willing to read the result if it combine charts and data together.

The image upload was an interesting panel, I'd rather than have to allow my entire gallery into the app just select a few images and allow them in like Instagram does

2.0 Interactive Prototype

Figma Prototype

- Use Figma to implement the basic wireframes
- Create low-fidelity interactive prototypes
- Implement navigation structure & core functionalities

Explore and consider different interface, interactions and design principles
(**Gestalt Principles of Perception and Norman Principles**)

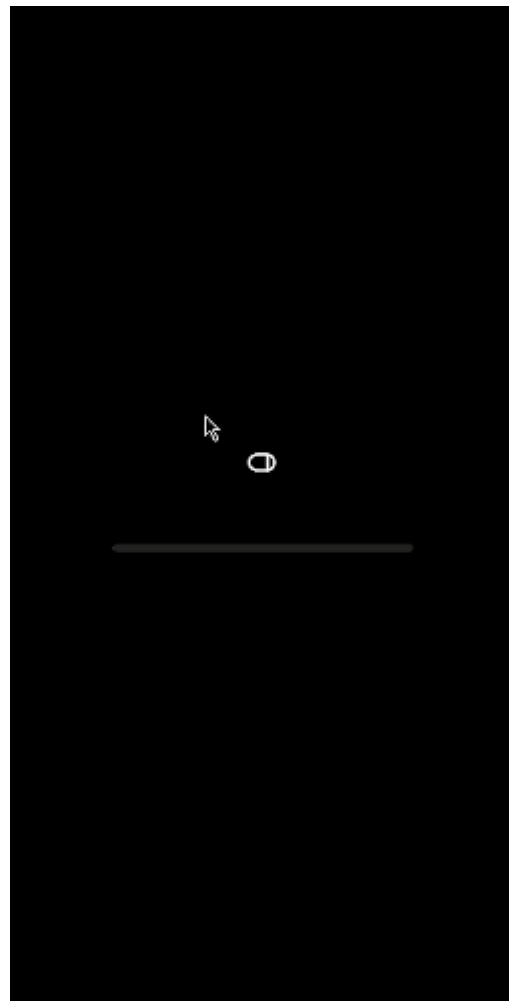
10. If you got stuck at any point, where did it happen?

2 responses

Just kidding, it just wasn't an option. The Resources tab freaked me out, it looks like one of those phishing scams where something looks just odd enough to catch me off guard. Maybe it's the amount of white space surrounding the text, or the scroll wheel just sitting there. It's like I opened a Kindle but I'm using a teeth app.

I did not

https://docs.google.com/forms/d/15v7dUTCo4fe_SbJqp7qeM3AN-971C8CMb7t_2fo8btY/edit#responses



2.0 Interactive Prototype Review

The **online survey + in person review**:

- Liked the Paper Prototype layout better
- Currently looks too sketchy and old -> not trustworthy
- Want a better theme colour

Any other suggestions

2 responses

I will more willing to read the result if it combine charts and data together.

The image upload was an interesting panel, I'd rather than have to allow my entire gallery into the app just select a few images and allow them in like Instagram does

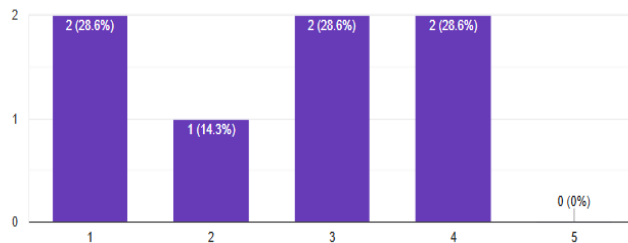
A visible outline or perhaps keeping the Home screen options nearby would make it feel nicer on the Resources page. Maybe an arrow on the top right with 'Back' and maybe on the History side on the top left showing 'Back' indicating the 3 pages nicely enough

17. How likely are you to use this app in real life?

(1 = Very unlikely, 5 = Very likely)

[Copy chart](#)

7 responses

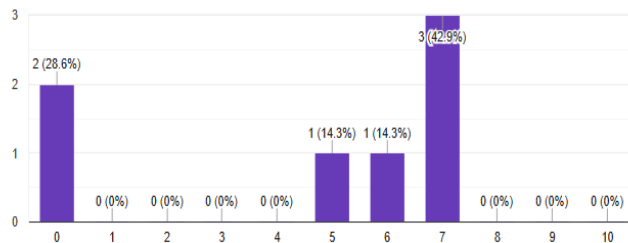


18. How likely are you to recommend this app to others?

(0 = Not at all likely, 10 = Extremely likely)

[Copy chart](#)

7 responses



3.0 Frontend + API Integration

Architecture Design

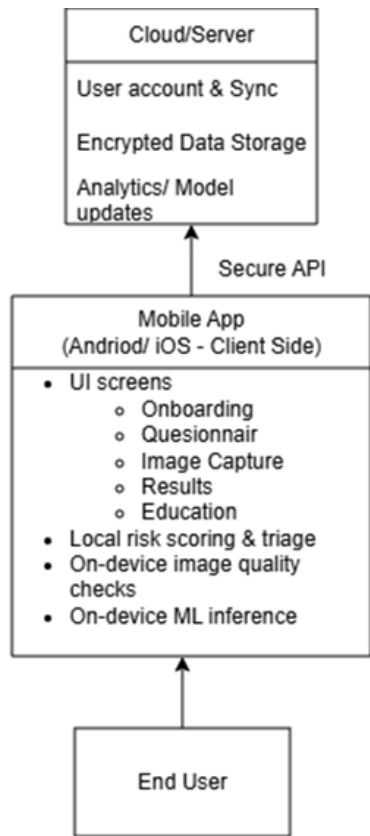
Mobile-First Design with an overall **client-server** architecture

This combination enables a clean client–server architecture where the frontend handles user interaction and the backend performs scalable AI-driven analysis.

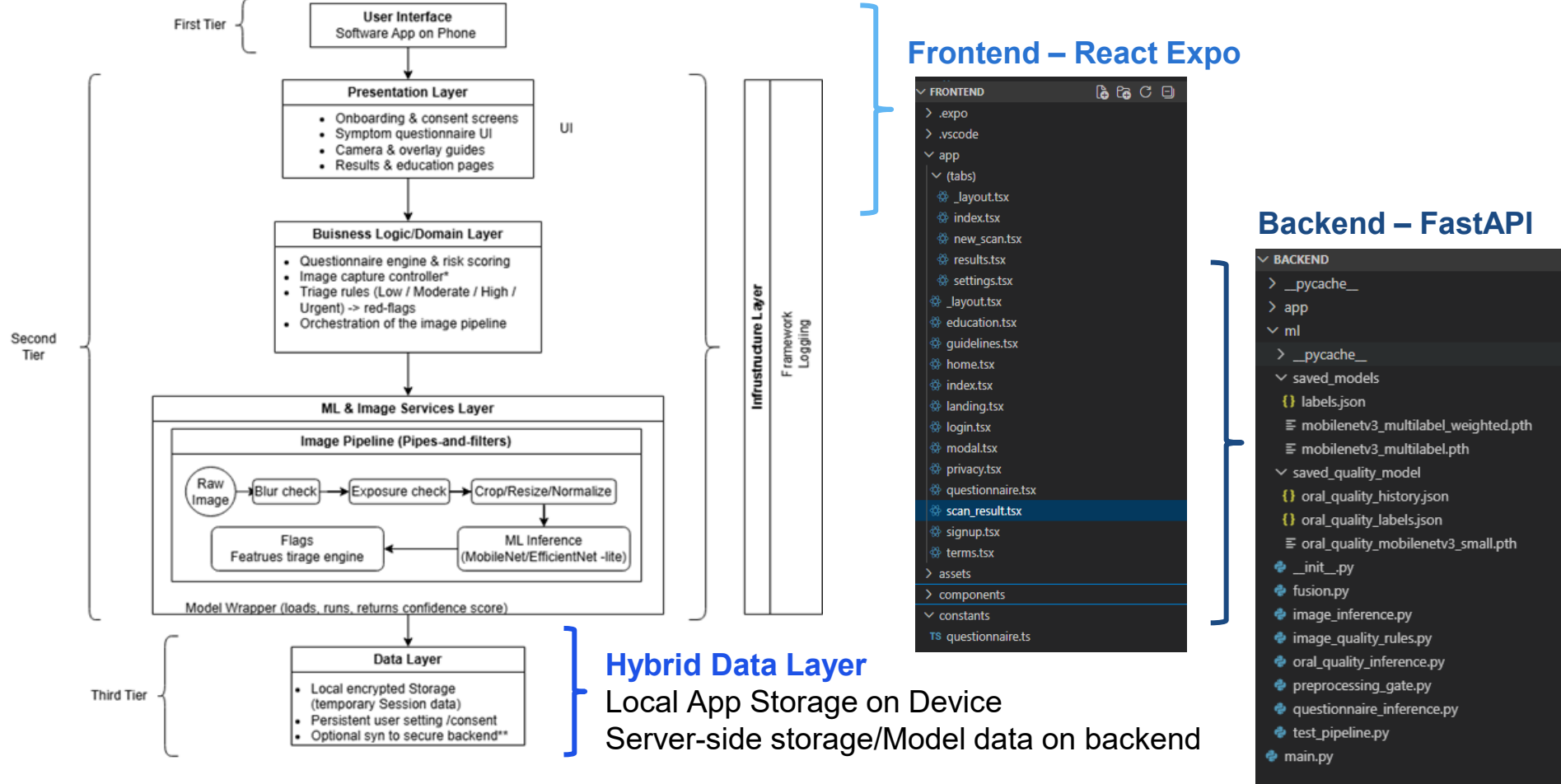
React (JS/TS) → sends HTTP requests

FastAPI → exposes REST endpoints (URL that backend calls)

Frontend - React Expo	Backend – FastAPI
• Handles UI/UX • User Interaction • Image Capture Native (Swift/Kotlin) → too slow for prototyping Flutter → less aligned with TypeScript stack	• Built for python • Image Preprocessing & Inference • Oral Inference • Questionnaire Inference • Fusion Node.js → awkward for ML Django → heavier, slower to prototype APIs



*System Architecture
Client-Server View.*



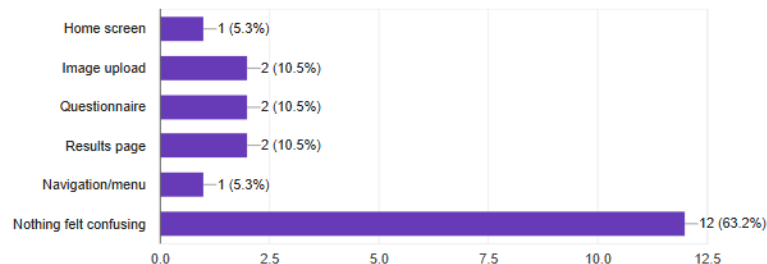
The project follows the same layered structure conceptually, but in implementation the ML & image services layer is deployed in the backend server rather than fully on-device.

3.0 Frontend + API Integration

9. Which part of the app felt most confusing, if any?

19 responses

[Copy chart](#)



10. If you got stuck at any point, where did it happen?

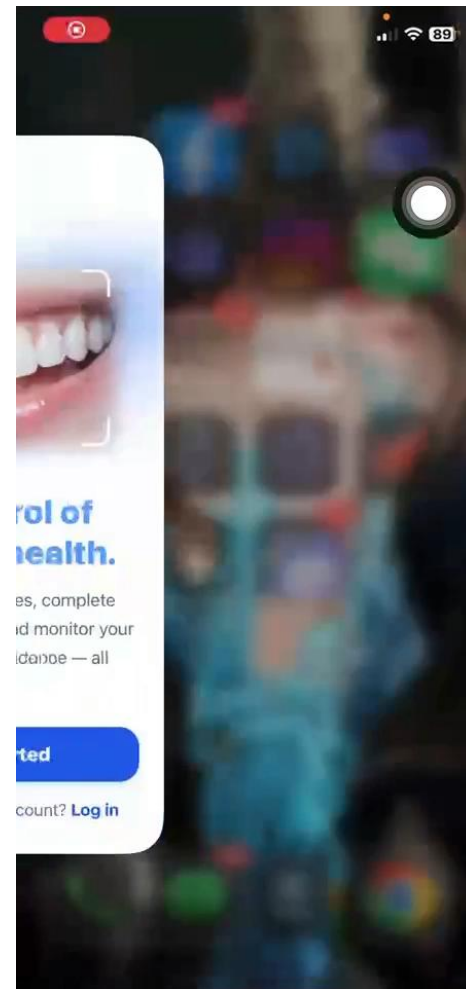
6 responses

N/A

Not stuck but I'm an Android user, I think an easier Delete button for past scans/images uploaded would be nice. Never used an iPhone so seeing the slide to delete was a first.

Why was a questionnaire required every time a scan was being done? it seems a bit off putting and like more work than necessary

User Interface

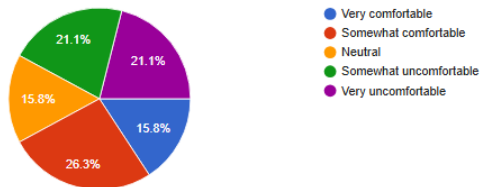


3.0 Frontend + API Integration

11. How comfortable would you feel uploading a photo of your teeth to this app?

 Copy chart

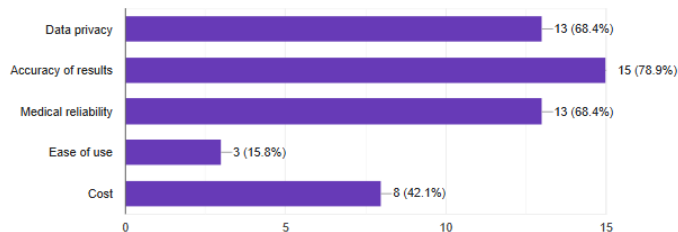
19 responses



12. What concerns, if any, do you have about using this app? (Select all that apply)

 Copy chart

19 responses

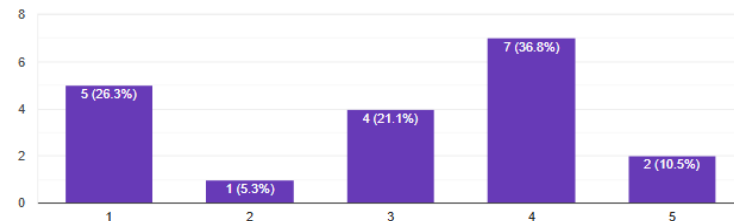


17. How likely are you to use this app in real life?

 Copy chart

(1 = Very unlikely, 5 = Very likely)

19 responses

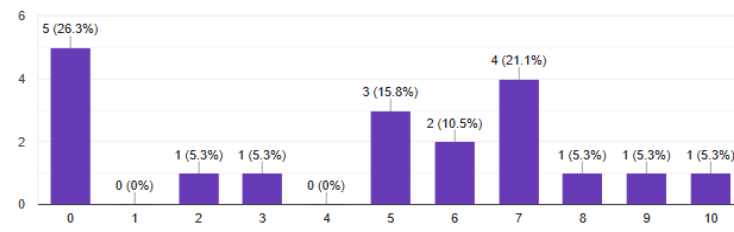


18. How likely are you to recommend this app to others?

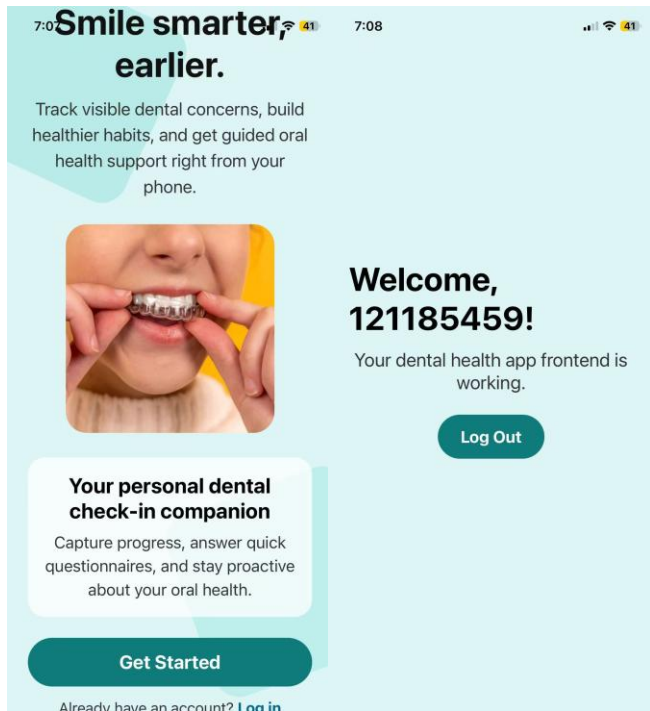
 Copy chart

(0 = Not at all likely, 10 = Extremely likely)

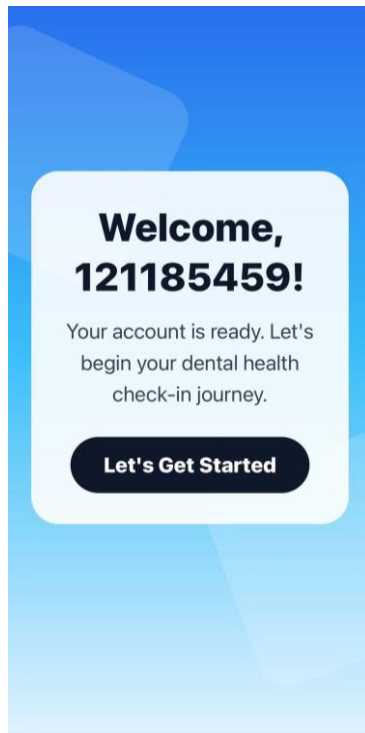
19 responses



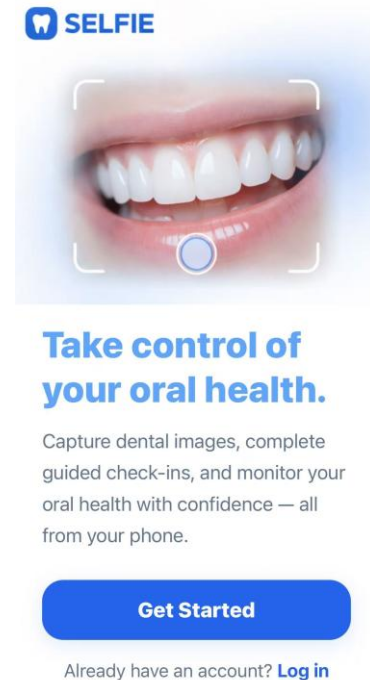
3.0 Frontend + API Integration Early Iterations



Early Prototypes of Landing & Welcome Page

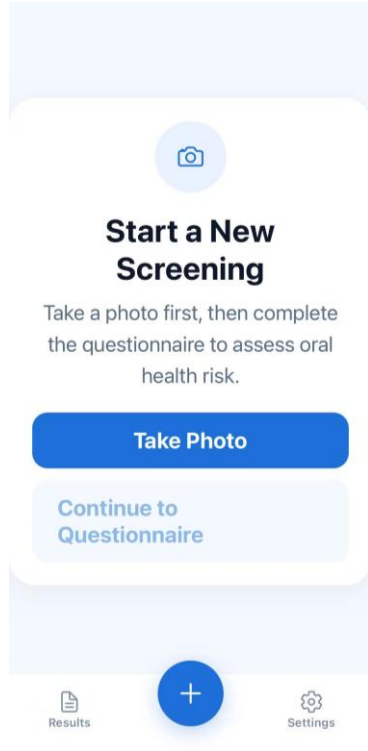


Experimenting Cards

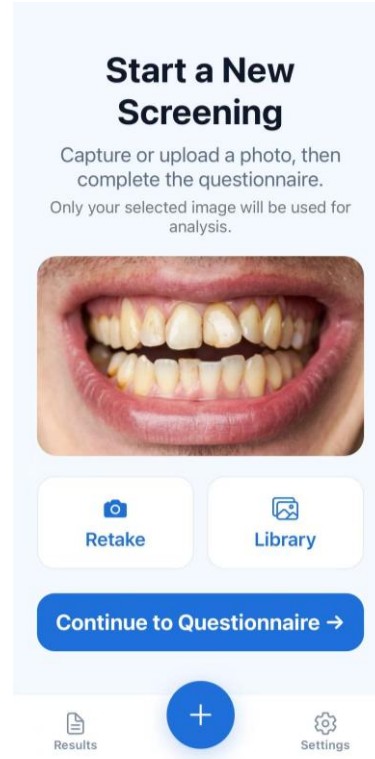


Professional Design

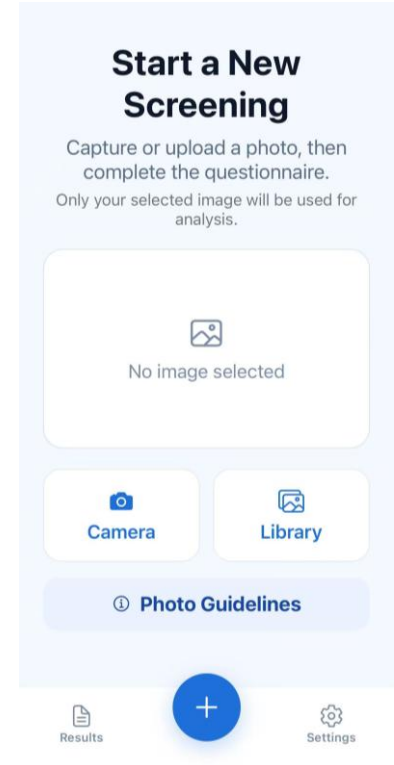
3.X Frontend + API Integration Early Iterations



3.0 Stakeholder want to upload photo directly from album



3.1 Need Guidelines for compliance and mitigation



3.2 Current Design

Implementing feedback from the Integration Demo

Past Scan Results

Detected: Tooth Discoloration, Calculus, Gingivitis, Caries

Image quality: pass

Model Probabilities

Tooth Discoloration: 64.9%

Calculus: 70.1%

Gingivitis: 77.3%

Caries: 88.5%

Questionnaire Answers

age_group: 35_to_55

dental_visits: 6-12mo

hygiene_habits: rarely

diet_sugar: rarely

staining_habits: diet_only

dry_mouth: yes

systemic_health: yes

trauma_history: yes

bleeding_gums: often

pain_sensitivity: mild



Results



Settings

Past Scan Results

Analysis completed successfully for oral_scan.jpg.

Detected: Tooth Discoloration, Calculus, Gingivitis, Caries

Image quality: pass

Risk Summary

● Low ● Moderate ● High

Caries: 88.5% • High Risk

Gingivitis: 77.3% • High Risk

Calculus: 70.1% • Moderate Risk

Tooth Discoloration: 64.9% • Moderate Risk

Recommendations

Caries — 88.5%

Reduce sugary and acidic foods, and schedule a dental exam to assess for possible cavities.

Gingivitis — 77.3%

Brush twice daily, floss every night, and consider a gum-care mouthwash. If



Results



Settings

Implemented Feedback:

1. Color Code the Risks
2. Remove the Demo features
3. Remove Image Status
4. Instead of saving questionnaire answers, save recommendations
5. Add time and date stamp of the scan
6. Want more visuals instead of crowded text
7. Being able to see the key results clearly
8. Having invalid images does not need to repeat the questionnaire before resubmission

Why was a questionnaire required every time a scan was being done? it seems a bit off putting and like more work than necessary

Implementing feedback from the Integration Demo

Past Scan Results



31 Tap to expand

Analysis completed successfully for oral_scan.jpg.

Detected: Tooth Discoloration, Gingivitis, Caries

Quality: pass



2026-03-31

Tap to expand

Analysis completed successfully



Results



Settings

Past Scan Results



2026-04-01, 1:30 AM

Tap to expand

Analysis completed successfully for oral_scan.jpg.

Detected: Healthy



2026-04-01, 1:03 AM

Tap to expand

Analysis completed successfully for oral_scan.jpg.

Detected: Calculus, Gingivitis, Teeth



Results



Settings

Implemented Feedback:

1. Color Code the Risks
2. Remove the Demo features
3. Remove Image Status
4. Instead of saving questionnaire answers, save recommendations
5. Add time and date stamp of the scan
6. Want more visuals instead of crowded text
7. Being able to see the key results clearly
8. Having invalid images does not need to repeat the questionnaire before resubmission

10. If you got stuck at any point, where did it happen?

6 responses

N/A

Not stuck but I'm an Android user, I think an easier Delete button for past scans/images uploaded would be nice. Never used an iPhone so seeing the slide to delete was a first.

Implementing feedback from the Integration Demo

Risk Overview

- Low (<50%)
- Moderate (50–74%)
- High (75%+)

Gingivitis 97.7% • High risk



Caries 78.7% • High risk



Calculus 76.9% • High risk



Tooth

Discolora 50.2% • Moderate risk
tion



Recommendations

Gingivitis — 97.7%

High likelihood of gingivitis. Your gums appear significantly red, swollen, or prone to bleeding.

Recommendation: Brush twice



Scan completed: 2026-03-31, 7:18 PM

Analysis completed successfully for oral_scan.jpg.

Detected: Gingivitis, Calculus, Caries

Risk Overview

- Low (<50%)
- Moderate (50–74%)
- High (75%+)

Gingivitis 97.2% • High risk



Tap to view details



Caries 79.2% • High risk



Scan completed: 2026-04-01, 1:31 AM

Risk Overview

- Low (<50%)
- Moderate (50–74%)
- High (75%+)

Gingivitis 94.3% • High risk



Tap to view details



Caries 72.3% • Moderate risk



Tap to view details



Calculus 67.7% • Moderate risk



Tap to view details



Feasibility Risk Analysis

Table 2. Key Failure Mode Names and Risk Priority Number

Key Failure Mode Index and Full Title	Risk Priority #	Mitigation Strategy
FM1 — False Reassurance / False Negative Triage	252	Model High Sensitivity
FM2 — Over-Triage / False Positive Triage	100	Fusion with Questionnaire Logic
FM3 — Improper Use / Incorrect Image Capture	175	Ask to retake/rechose photo
FM4 — Skipped or Inconsistent Questionnaire Responses	140	Block the user from progressing if questions are unanswered
FM5 — App Crash / Freeze / Unresponsive Behaviour	72	Error messages popping up
FM6 — Server Downtime or Slow Backend Response	72	Error messages popping up
FM7 — Algorithmic Bias / Unequal Performance Across Subgroups	144	Larger dataset, willful incorporation of available data from subgroups
FM8 — Data Breach / Unauthorized Access to Health Data	162	AES + TSL 1.3 Encryption
FM9 — Misuse or Unclear Ownership of Personal Data	120	Terms & Agreements; Privacy & Security
FM10 — Regulatory Non-Compliance	64	Terms & Agreements; Privacy & Security
FM11 — Over-Reliance on the App / Misunderstanding Scope of Use	140	Terms & Agreements; Privacy & Security

DO3 Verification Plan

DO3 Privacy, Consent, and Access Control Documentation Verification Plan

To verify DO3, the user interface of this project will be inspected against the following industry standards:

- Google Material Design [22]
- Microsoft Fluent Design System [23]
- WCAG 2.1 Accessibility Guidelines [24]

Due to the number amount of NFRs included in this section, the verifications and analyses will have an emphasis on UI performance with the following verification method and acceptance criteria:

- Interaction Latency Test: $\geq 95\%$ of UI interactions respond within ≤ 100 ms
 - Human perception is quantifiable as Google and Microsoft states delays above 100ms are noticeable [22] [23]
- Frame Rendering Analysis: $\geq 95\%$ of frames rendered at ≥ 60 fps
 - Industrial standard for smooth mobile rendering
- WCAG 2.1 AA compliance
 - Baseline of accessibility for mobile platforms [24]

RAM	JSC	Views	UI	JS
243.58	0.00	0	60	60
MB	MB	0		

DO3 Latency Testing

Main Test: Measure Screening Process Latency

Testing includes:

- Next & Option Navigation Latency
- Backend Request Latency
- Submit UI Latency

Next Step:

Identify Bottle Neck (image upload? Preprocessing? ML Inference?)

Ensure backend is always warmed up

Resize/Compress image if possible -> set lowest possible working slides

```
LOG Option selection latency: 29 ms
LOG Next navigation latency: 31 ms
LOG Option selection latency: 37 ms
LOG submitScan uri: file:///var/mobile/Containers/Data/Application/0F215607-38D1-4D62-AA41-26578D1AAC41/Library/Caches/ExponentExperienceData/@anonym
agePicker/727EE717-B83D-42EF-A3A5-DD345FC58110.jpg
LOG submitScan answers: {"age_group": "35 to 55", "bleeding_gums": "sometimes", "dental_visits": "6-12mo", "diet_sugar": "sometimes", "dry_mouth": "yes",
: "mild", "staining_habits": "tobacco", "systemic_health": "yes", "trauma_history": "no"}
LOG Sending request to: https://sfurhme-5p06.onrender.com/analyze
LOG Backend status: 200
LOG Raw backend response: {"success":true,"summary":"Analysis completed successfully for oral_scan.jpg.", "image_quality_passed":true, "image_quality_status
atus": "pass", "rules_report": {"status": "pass", "checks": {"resolution": {"passed": true, "width": 612, "height": 408, "message": null}, "brightness": {"passed": true, "me
, "too_bright": false, "message": null}, "blur": {"passed": true, "blur_score": 98.93493397255953, "message": null}, "contrast": {"passed": true, "contrast_score": 38.0229
s": []}, "model_report": {"passed": true, "predicted_class": "valid oral", "confidence": 0.9653937816619873, "message": null, "failures": [], "warnings": []}, "received
6-12mo", "hygiene_habits": "once_daily", "diet_sugar": "sometimes", "staining_habits": "tobacco", "dry_mouth": "yes", "systemic_health": "yes", "trauma_history": "no",
ld"}, "image_probabilities": {"Calculus": 0.99800822745132446, "Caries": 0.7880412936210632, "Gingivitis": 0.9940217137336731, "Tooth Discoloration": 0.5013675093650
775406687981454, "Caries": 0.9820137900379085, "Gingivitis": 0.9975273768433653, "Tooth Discoloration": 0.6224593312018546}, "probabilities": {"Tooth Discoloration
alculus": 0.77564071251296, "Gingivitis": 0.99542397897755}, "detected_conditions": ["Tooth Discoloration", "Caries", "Calculus", "Gingivitis"]}]
LOG Backend request latency: 4525 ms
LOG Total submit UI latency: 4639 ms
```

\\McMaster_University\6thYear_2025-20

```
Option selection latency: 32 ms
Next navigation latency: 45 ms
Option selection latency: 35 ms
Next navigation latency: 40 ms
Option selection latency: 22 ms
Next navigation latency: 31 ms
Option selection latency: 30 ms
Next navigation latency: 38 ms
Option selection latency: 27 ms
Next navigation latency: 30 ms
Option selection latency: 30 ms
Next navigation latency: 29 ms
Option selection latency: 25 ms
Next navigation latency: 32 ms
Option selection latency: 36 ms
Next navigation latency: 36 ms
Option selection latency: 29 ms
Next navigation latency: 31 ms
Option selection latency: 37 ms
```

Testing & Validation FM5/6

Error Message will pop up

- Logged on HTTPS website and error message displayed on phone

```
LOG submitScan answers: {"age_group": "under_35", "bleeding_gums": "sometimes", "dental_visits": "2mo", "diet_sugar": "rarely", "dry_mouth": "yes", "hygiene_habits": "rarely", "pain_sensitivity": "d", "staining_habits": "diet_only", "systemic_health": "yes", "trauma_history": "no"}
LOG Sending request to: https://sfwrhme-5p06.onrender.com//analyze
LOG Backend status: 404
LOG Raw backend response: {"detail": "Not Found"}
ERROR submitScan failed: [Error: Backend error: 404 {"detail": "Not Found"}]
```

all Stack

```
submitScan (services\scanService.ts)
next (<native>)
asyncGeneratorStep (node_modules\@babel\runtime\helpers\asyncToGenerator.js)
_next (node_modules\@babel\runtime\helpers\asyncToGenerator.js)
tryCallOne (address at (InternalBytecode.js:1:1180))
anonymous (address at (InternalBytecode.js:1:1874))
ERROR Questionnaire submit failed: [Error: Backend error: 404 {"detail": "Not Found"}]
```

Questionnaire

Answer each question as accurately as possible to support your oral health screening.

Question 10 of 10

100%

Submission failed

We could not submit your questionnaire right now. Please check your connection and try again.

OK



cold/sweets



Severe, constant pain or pain when chewing

2

Questionnaire submit failed: E... ✕

DO4 Testing & FM8 Mitigation

Implementation of TLS 1.2+

- OWASP recommends secure REST services expose on to HTTPS endpoints
- TLS protects credentials & transmitted data in transit
- Ideal for the frontend to backend traffic
- AES-GCM will be added in later implementations for data at rest
 - Currently not necessary due to local database for user data

DO4 – Data Schema & Encryption Documentation Verification Plan

To verify DO4, the method of encryption must be industry-approved cryptographic algorithms such as AES-256 or TLS 1.2+. 100% of sensitive data fields must be encrypted at rest and in transit to ensure meeting encryption standards, further preventing unauthorized data disclosure and mitigating breach risks.

Back End Testing

Creating a mock_backend.py file to see if the data could be received

- A fake backend that simulates Responses without running the ML

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\McMaster_University\6thYear_2025-2026\SFWRBME_5P06\SFWRBME-5P06\Capstone\frontend\Tests> python mock_backend.py
```

```
PS E:\McMaster_University\6thYear_2025-2026\SFWRBME_5P06\SFWRBME-5P06\Capstone\frontend\Tests> python mock_backend.py
```

```
INFO: Started server process [27828]
```

```
INFO: Waiting for application startup.
```

```
INFO: Application startup complete.
```

```
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

```
=== MOCK BACKEND RECEIVED ===
```

```
filename: oral_scan.jpg
```

```
content_type: image/jpeg
```

```
image_size_bytes: 2654434
```

```
answers: {'missing_teeth': 'none', 'last_cleaning': '<6mo', 'bleeding_gums': 'never', 'red_swollen_gums': 'no', 'loose_teeth': 'no', 'hard_deposits': 'no', 'pain_sensitivity': 'no', 'visible_holes': 'no', 'bad_breath': 'no', 'age_group': '<18'}
```

```
INFO: 192.168.1.11:53014 - "POST /analyze HTTP/1.1" 200 OK
```

```
[]
```

Back End Testing

Backend Created with HTTP using local IP

- <http://127.0.0.1:8000/docs> showing that the data is received and analyzed by the ML

```
Unexpected key(s) in state_dict: 'Classifier.3.0.weight', 'Classifier.3.0.bias', 'Classifier.3.3.weight', 'Classifier.3.3.bias'.
WARNING: StatReload detected changes in 'ml\oral_quality_inference.py'. Reloading...
INFO: Started server process [30936]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [30936]
INFO: Stopping reloader process [5272]
PS E:\McMaster_University\6thYear_2025-2026\SFWRBME_5P06\SFWRBME-5P06\Capstone\backend> python -m uvicorn main:app --reload --host 0.0.0.0 --port 8000
INFO: Will watch for changes in these directories: ['E:\McMaster_University\6thYear_2025-2026\SFWRBME_5P06\SFWRBME-5P06\Capstone\backend']
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Started reloader process [37260] using StatReload
INFO: Started server process [33816]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:11331 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:11331 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 192.168.1.11:53470 - "POST /analyze HTTP/1.1" 200 OK
```

Back End Testing

Backend Created with HTTPS using local IP

- <http://127.0.0.1:8000/docs> showing that the data is received and analyzed by the ML (old survey 1)

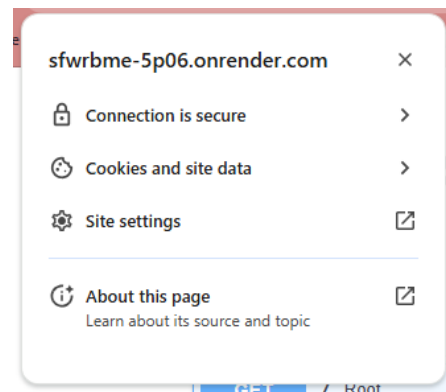
```
C:\Users\RUIDIL~1\AppData\Local\Temp\tmpts6ubcgd.jpg
INFO:      192.168.1.11:53859 - "POST /analyze HTTP/1.1" 200 OK

=== IMAGE RECEIVED ===
Filename: oral_scan.jpg
Bytes received: 51789
=== ANSWERS RECEIVED ===
{'age_group': 'under_35', 'dental_visits': '<6mo', 'hygiene_habits': 'twice_daily', 'diet_sugar': 'rarely', 'staining_habits': 'no', 'dry_mouth': 'no', 'systemic_health': 'no', 'trauma_history': 'no', 'bleeding_gums': 'never', 'pain_sensitivity': 'no'}
=== TEMP IMAGE SAVED ===
Temp path: C:\Users\RUIDIL~1\AppData\Local\Temp\tmpqgrj_u86.jpg
=== PRECHECK RESULT ===
{'status': 'pass', 'rules_report': {'status': 'pass', 'checks': {'resolution': {'passed': True, 'width': 612, 'height': 408, 'message': None}, 'brightness': {'passed': True, 'mean_brightness': 142.18550557477894, 'too_dark': False, 'too_bright': False, 'message': None}, 'blur': {'passed': True, 'blur_score': 127.80505754242, 'message': None}, 'contrast': {'passed': True, 'contrast_score': 51.857365379679266, 'message': None}}, 'failures': [], 'warnings': [], 'model_report': {'passed': True, 'predicted_class': 'valid_oral', 'confidence': 0.9698722958564758, 'message': None, 'failures': [], 'warnings': []}}
=== IMAGE INFERENCE RESULT ===
{'Calculus': 0.8746572732925415, 'Caries': 0.8428335189819336, 'Gingivitis': 0.6220372915267944, 'Tooth Discoloration': 0.5474687814712524}
=== QUESTIONNAIRE INFERENCE RESULT ===
{'Calculus': 0.05, 'Caries': 0.05, 'Gingivitis': 0.05, 'Tooth Discoloration': 0.05}
=== DETECTED CONDITIONS (TEMP: IMAGE ONLY) ===
['Calculus', 'Caries', 'Gingivitis', 'Tooth Discoloration']
=== TEMP FILE REMOVED ===
C:\Users\RUIDIL~1\AppData\Local\Temp\tmpqgrj_u86.jpg
INFO:      192.168.1.11:53859 - "POST /analyze HTTP/1.1" 200 OK
```

Testing & Validation FM8

Testing of TLS

- We currently use Render for TLS which automatically provides certificates <https://sfwrbme-5p06.onrender.com/> HTTPS = HTTP over TLS
 - **Verification 1:** <https://sfwrbme-5p06.onrender.com/docs>
 - **Verification 2:** Curl.exe a command-line tool that sends HTTP/HTTPS requests
 - **Verification 3:** Logs on Render



```
PS C:\Users\Ruidi Liu> curl.exe -v https://sfwrbme-5p06.onrender.com/health
* Host sfwrbme-5p06.onrender.com:443 was resolved.
* IPv6: (none)
* IPv4: 216.24.57.251, 216.24.57.7
* Trying 216.24.57.251:443...
* schannel: disabled automatic use of client certificate
* ALPN: curl offers http/1.1
* ALPN: server accepted http/1.1
* Established connection to sfwrbme-5p06.onrender.com (216.24.57.251 port 443) from 192.168.1.13 port 14119
* using HTTP/1.x
> GET /health HTTP/1.1
> Host: sfwrbme-5p06.onrender.com
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
* schannel: remote party requests renegotiation
* schannel: renegotiating SSL/TLS connection
* schannel: SSL/TLS connection renegotiated
< HTTP/1.1 200 OK
< Date: Wed, 01 Apr 2026 16:38:48 GMT
< Content-Type: application/json
< Transfer-Encoding: chunked
< Connection: keep-alive
< rndr-id: 87e3fced-9e95-4f38
< vary: Accept-Encoding
< x-render-origin-server: uvicorn
< cf-cache-status: DYNAMIC
< Server: cloudflare
< CF-RAY: 9e5906b61df839d2-YYZ
< alt-svc: h3=":443"; ma=86400
<
{"status":"ok"}* Connection #0 to host sfwrbme-5p06.onrender.com:443 left intact
PS C:\Users\Ruidi Liu>
```

dashboards.render.com/web/srv-d76je8muk2gs73f13el0/logs?r=live

Online SchoolingFFXIVHousingDrawingJobApplication

My Workspace

My project / Production / SFWRBME-5P06

Search

+ NewUpgrade

Environment

SFWRBME-5P06

Events

Settings

MONITOR

Logs

Metrics

MANAGE

Environment

Shell

Scaling

Previews

Disk

Jobs

Workflows: Now in beta

Orchestrate your agents and background jobs with automatic retries and scale-to-zero. [Learn more](#)

Create Workflow

Changelog

Invite a friend

Contact support

Render Status

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

All logsSearch

Live tail

Apr 1

12:16:22 PM ==>

12:16:22 PM ==> //////////////////////////////////

12:16:22 PM ==>

12:16:22 PM ==> Available at your primary URL <https://sfwrbme-5p06.onrender.com>

12:16:22 PM ==>

12:16:22 PM ==> //////////////////////////////////

12:16:22 PM [k65sq] INFO: 34.82.79.105:0 - "GET / HTTP/1.1" 200 OK

12:16:25 PM [k65sq] INFO: 216.165.204.245:0 - "HEAD /docs HTTP/1.1" 200 OK

12:16:25 PM [k65sq] INFO: 216.165.204.245:0 - "GET /docs HTTP/1.1" 200 OK

12:16:25 PM [k65sq] INFO: 216.165.204.245:0 - "GET /openapi.json HTTP/1.1" 200 OK

12:17:23 PM [k65sq] INFO: 216.165.204.245:0 - "GET / HTTP/1.1" 200 OK

12:17:29 PM [k65sq] INFO: 216.165.204.245:0 - "GET /docs HTTP/1.1" 200 OK

12:17:29 PM [k65sq] INFO: 216.165.204.245:0 - "GET /openapi.json HTTP/1.1" 200 OK

12:21:11 PM [k65sq]

12:21:11 PM [k65sq] === /analyze hit ===

12:21:11 PM [k65sq] === IMAGE RECEIVED ===

12:21:11 PM [k65sq] Filename: oral_scan.jpg

12:21:11 PM [k65sq] Content type: image/jpeg

12:21:11 PM [k65sq] Bytes received: 48219

12:21:11 PM [k65sq] === ANSWERS RECEIVED ===

12:21:11 PM [k65sq] {'age_group': '35_to_55', 'dental_visits': '0-12mo', 'hygiene_habits': 'once_daily', 'diet_sugar': 'sometimes', 'staining_habits': 'diet_only', 'dry_mouth': 'yes', 'systemic_health': 'yes', 'trauma_history': 'yes', 'bleeding_gums': 'sometimes', 'pain_sensitivity': 'mild'}

12:21:11 PM [k65sq] === TEMP IMAGE SAVED ===

12:21:11 PM [k65sq] Temp path: /tmp/tmpk_jw3j6w.jpg

12:21:11 PM [k65sq] === PREGHECK RESULT ===

12:21:11 PM [k65sq] {'status': 'pass', 'rules_report': {'status': 'pass', 'checks': {'resolution': {'passed': True, 'width': 612, 'height': 408, 'message': None}, 'brightness': {'passed': True, 'mean_brightness': 145.2125664808407, 'too_dark': False, 'too_bright': False, 'message': None}, 'blur': {'passed': True, 'blur_score': 164.29759800143043, 'message': None}, 'contrast': {'passed': True, 'contrast_score': 57.02357702683753, 'message': None}}, 'failures': [], 'warnings': [], 'model_report': {'passed': True, 'predicted_class': 'valid_oral', 'confidence': 0.9878573417663574, 'message': None}, 'failures': [], 'warnings': []}}

12:21:11 PM [k65sq] === IMAGE INFERENCE RESULT ===

12:21:11 PM [k65sq] {'Calculus': 0.9425749778747559, 'Caries': 0.6174694299697876, 'Gingivitis': 0.9067448973655761, 'Tooth Discoloration': 0.7543359398841858}

12:21:11 PM [k65sq] === QUESTIONNAIRE INFERENCE RESULT ===

12:21:11 PM [k65sq] {'Calculus': 0.1192629220221755, 'Caries': 0.9826137908379085, 'Gingivitis': 0.9525741268224334, 'Tooth Discoloration': 0.9525741268224334}

12:21:11 PM [k65sq] === FUSION RESULT ===

12:21:11 PM [k65sq] {'Tooth Discoloration': 0.8038954866187477, 'Caries': 0.7268327379902239, 'Calculus': 0.6543947583263324, 'Gingivitis': 0.9256765891483155}

12:21:11 PM [k65sq] === DETECTED CONDITIONS ===

12:21:11 PM [k65sq] ['Tooth Discoloration', 'Caries', 'Calculus', 'Gingivitis']

12:21:11 PM [k65sq] === TEMP FILE REMOVED ===

12:21:11 PM [k65sq] /tmp/tmpk_jw3j6w.jpg

12:21:11 PM [k65sq] INFO: 216.165.204.245:0 - "POST /analyze HTTP/1.1" 200 OK

12:21:29 PM ==> Detected service running on port 10000

12:21:29 PM ==> Docs on specifying a port: <https://render.com/docs/web-services#port-binding>

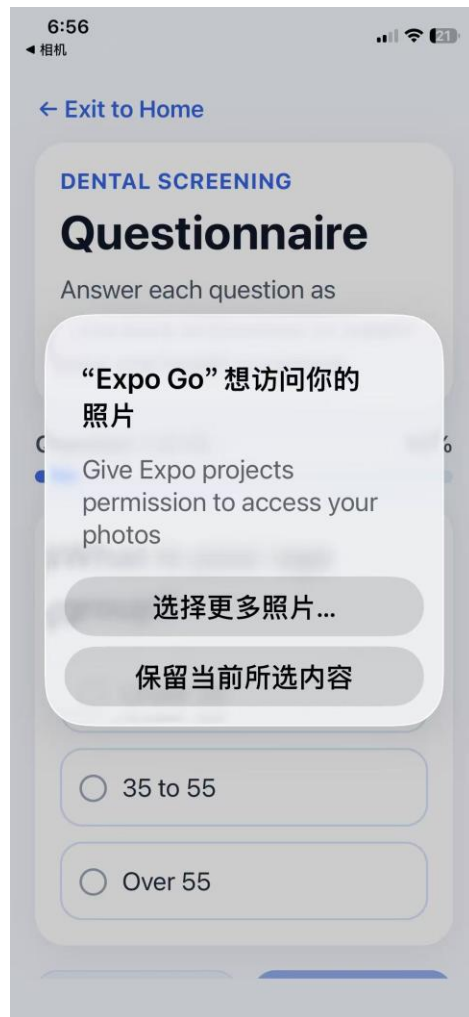
Further Integration

iOS Photo Library Permission Behavior

- When users select an image from the photo library, iOS displays a system-level prompt: “Select More Photos” / “Keep Current Selection”
- This occurs due to iOS limited photo library access permissions
- The issue persists even after removing redundant permission requests in code
- Root cause: OS-level behavior outside application control

Next Step:

- Investigate native configuration options within Expo (React Native framework) or iOS settings to better manage or minimize the prompt experience



Further Integration

Cloud-Based Database Integration

Current implementation stores user data locally

Limits:

- No cross-device access
- No persistent user profiles
- No scalability for real-world deployment

Next Step: Migrate to a cloud-hosted database to support:

- Secure user account storage
- Authentication & session management
- Scalable data handling

Legal & Compliance Improvements

Current system includes Terms & Conditions acknowledgment

Missing: Timestamped consent records

Next Step:

- Store timestamp of user agreement
- Store Version of terms accepted

Ensures

- Auditability
- Legal traceability
- Alignment with privacy standards

Further Integration

Accessibility & Comfort Design Enhancements

Feedback from user interviews highlighted accessibility needs:

- Color vision deficiencies
- Preference for dark mode

Next Step: Implement:

- Color-blind friendly palettes (WCAG-aligned contrast)
- Voice assistant/support
- Dark mode support
- Improves usability, inclusivity, and user trust
- **MUST align with WCAG 2.1 AA compliance**

Thank You!

Generalized Design Inputs & Outputs

Table 3. Design input and corresponding Design Output

Design Input	Objective & Measurable Description	Design Output
Accessibility	- meet WCAG 2.1 AA standards for all user-facing interfaces [6]	UI/UX Wireframes & Specifications
Affordability	- run on widely available smartphones (Android 8+/iOS 13+) [7][8] - no external hardware	Supported Device List
Usability	- ≥ 90% task completion on first attempt [11] - Complete full screening workflow in ≤ 5 minutes on average [12].	Usability Performance Output
Extensibility	Support integration of ≥ 1 additional diagnostic or screening module without modifying core triage logic [13] [14].	System Architecture Output
Privacy & Security	- AES-256 encryption used for personal health data [15] - TLS 1.2+ .used for communication [15]	Data Storage & Encryption Specification [16]
Deterministic Triage	- Triage logic only produce 4 defined outputs [17] - achieve ≥ 95% branch test coverage [18][19]	Triage Recommendation Module Output

FMEA Spreadsheet Template

System / Product / Process		FMEA ID	Revision	Date
Selfie Dental Assistant Application				Dec. 1st 2025
Subsystem / Product / Process (if applicable)		Party Responsible	Approved By	Date Completed
		Team 9		
Team				
Team 9 Ruid Liu & Jasmine Wong - Current Version 1.0				

Table 3. FMEA Table

Failure Mode and Effects Analysis (FMEA)

FMEA Results																						
ID	System / Item / Process Step	Subsystems	Function	Potential Failure Mode	Effects of Failure	Severity	Causes	Occurrence	Current Controls	Prevention	Detection	Detection	RPN	Recommended Actions	Owner	Date Due	Action Results	Score Complete at	Severity	Occurrence	Detection	RPN
#	Name, ID number, etc.	What subsystem does this failure mode belong to?	Primary function	How could the system / item / process potentially fail?	Consequential impact on other systems, departments, etc.	1-10	All contributing factors	1-10	How to prevent the failure from occurring?	How to detect the failure from occurring?	1-10	1-10	1-10	Steps required to reduce severity, occurrence, and detection	Organization, team, or individual responsible	Target date of completion	Actions taken	Actual date of completion	1-10	1-10	1-10	1-10
###	RND: Home connection / Slow negative flow	Business Logic / Domain Layer	Perform accurate bid scoring and stage decision-making by combining customer resources, image analysis model, and real-time rules.	Default condition classified as low risk or fail due to number of items involved per function	Delayed case progression of disease, increased pain, potential loss of a limb or other body structure	9	Incomplete or inaccurate symptoms, poor follow-up of customer service, inadequate training, high long-term clinical costs, etc.	4	Real-time override rules, mandatory key questions that cannot be skipped, conservative high-specific thresholds and test on image logic. Clinical has higher weight than facility. Provide suggestion to see a doctor to monitor what the machine	Clinical validation against labelled cases; monitoring of huge distribution of other patients	7	252	Develop and implement a VAPT and model deployment. Upgrade image output trends. Flag when error occurs along beyond data set	Jasmine	Mid February to March						0	
###	RND: Over-Image / Slow positive flow	ML Engine Service Layer	Process user images through the image pipeline (data check -> exposure check -> compression/formatting) and perform ML inference using standardized thresholded data	benign findings classified as high risk / urgent	Unnecessary anxiety and appointments, increased system burden	5	Lighting artifacts, effective software, overly conservative thresholds	5	Pre-processing (data response normalization), conservative wording of results, non-emergency wording	Review of confusion matrix in validation and plot monitor high-risk rate in production	4	100	Relative testing & adjust threshold, expand beyond data set	Jasmine	March						0	
###	RND: Image crop / Increased image quality	Presentation Layer	Support user in capturing high-quality informed photographs using guided overlays, instructions, and UI prompts that enhance correct image framing.	User takes poor quality or blurred images	Inaccurate ML features, potential false reassurance or false claims	7	User confusion, low health literacy, lighting, framing, patient handling, camera shake or lens flare	5	Step-by-step camera instructions overlay guidelines, enhanced bid response checks, plain language prompts	Apparent quality scores analysis on failed images and check images before passing to model	5	175	Perform data-driven iterations: Outline client feedback and update control guidelines. Highlight and check images before passing to model	Ruid	Mid February						0	
###	RND: Stopped or inconsistent questionnaire responses	Presentation Business Logic Layer	Collect accurate and complete symptom and history data from the user and pass the structured responses to the ML scoring engine	Mistake or contradictory symptom data used for bid scoring	Inaccurate bid assessment, potential under- or over-triage	7	Data entry, misinterpreted questions, accidental click on irrelevant choices	4	Mandatory responses for high-impact items, confirmation for extreme values, simple consistency checks	Focused analysis on misaligned/changed responses; flag when internal checks trigger	5	140	Review and confirm that answers follow user to have a method to re-answer questions for all questions on misalignment	Ruid	Mid February to March						0	
###	RND: App crash / Error / Unresponsive behaviour	Infrastructure Layer	Provide a stable, responsive runtime environment for the mobile app, including UI rendering, ML model loading, memory management, and background processes	App stops responding during assessment	Interrupted assessment, user frustration, possible abandonment and delayed care	6	OS/SDK incompatibility, memory leaks, unhandled exceptions, heavy model on low-end device	4	Automatic guarded model loading, if/except error handling, device/CPU support notifications, low mode	Crash analysis and logs from prod, user feedback reports	3	72	Optimize ML model using quantization and model size reduction; expand device compatibility testing; build automated crash reporting and analytics; ensure application stability across every step	Both members	April						0	
###	RND: Server downtime / Slow backend response	Cloud / Server Layer	Enable secure data synchronization, storage, model deployment, and user account services through a reliable, scalable, accessible, and encrypted APIs	Backend unavailable or very slow	Images not uploaded, results delayed or absent, incomplete assessments	6	Network outages, server misconfiguration, untested, outdated certificates	3	Graceful error messages, retry with backoff, symptom-only on device fallback	Server uptime monitoring, API latency dashboards	4	72	Provide fully offline questionnaire fallback if server is unavailable; perform load testing to identify bottlenecks; upgrade model capacity	Both members	March						0	
###	RND: Algorithmic bias / Performance across subgroups	ML Engine Service Layer + Cloud	Monitor and update the machine learning model, ensuring that inference pipeline remains accurate, fit, and up-to-date across user populations and devices	Systematically worse performance for some demographics or device groups	Inequitable care, higher false negative or positive for certain users	8	Non-representative training data, biased labels, correlated subgroup performance	3	Device testing, active bias metrics by subgroup, clinician oversight on high-risk cases	Periodic bias audit on validation and pilot data; monitoring of subgroup performance metrics	6	144	Perform periodic bias audits; expand dataset with intentionally diverse images (skin tone, age, camera type); Add fairness evaluation metrics to model metrics; perform audits before model release; Tune fairness-related hyperparameters for groups if systematic patterns emerge in model outputs	Jasmine	April						0	
###	RND: Data breach / Unauthorized access to health data	Data Layer (Server) + Cloud / Server Layer	Store and protect personal health information (PHI) - including images, questionnaire responses, and consent settings - using encryption, access controls, and secure proper collection, use, retention, and ownership of user data, including maintaining correct records and applying privacy-based data governance rules	PHI or images accessed by unauthorized party	Privacy violations, possible identity risk, regulatory penalties, loss of trust	9	Weak encryption, insecure storage, misconfigured access, inadequate access control	3	Encryption of data and in transit, least privilege access, secure CD storage, audit logging, data minimization	Security monitoring, log review, intrusion/detection alerts	6	162	Strengthen encryption (AES-256 for local storage, enhanced TLS 1.3) for transit; conduct regular reviewing of access to PHI	Ruid	Late March						0	
###	RND: Abuse or misuse / Compromise of personal data	Data Layer + Regulatory & Compliance Process	Ensure proper collection, use, retention, and ownership of user data, including maintaining correct records and applying privacy-based data governance rules	Data used for lit. or shared without meaningful consent	Unlawful processing, user distress, legal / regulatory risk	8	Analytics consent (not) being, data sharing, permissions, long retention	3	Explicit consent form, separate opt-in for research/sharing, documented retention limits, transparent policy	Regular policy reviews, internal audits of data use in consent records	5	120	Revisit consent screen in plain language with explicit opt-in for data use in ML; add in-trip privacy dashboard or note showing what data is stored and why; implement data deletion on user request	Both members	February						0	
###	RND: Regulatory non-compliance	System-wide / Regulatory & Compliance Process	Monitor compliance with regulatory standards and codes (CCAC/CDNA, HIPAA, NPSA, CAD, S&D, cybersecurity guidelines) through documentation, quality management, and controlled updates	Misalignment with CCAC/CDNA, S&D, privacy, CAD, or cybersecurity guidance	Legal penalties, forced product withdrawal, reputational damage	8	Incomplete regulatory analysis, missing documentation, ad hoc updates	2	Map requirements to follow secure development guidance	Internal compliance review, external consultation if needed	4	64	Prepare third-party audit management (e.g. SOC 1/2/3) template; maintain design history and software lifecycle documentation; Review data flows to maintain compliance with HIPAA/HITECH	Both members	April						0	
###	RND: Over-reliance on app / Misunderstanding advice	Presentation Layer (Health & education pages)	Communicate bid results, explanations, and educational guidance to client, non-diagnostic language to support user understanding and promote appropriate follow-up care	User feels no result or defective diagnosis and delays seeing a doctor	Delay in needed care, worsening of condition	7	Weak or hidden disclaimers, over-reliance on AI, limited health literacy	4	Provided "not a diagnosis" messaging; include check-up recommendation even for low-risk; clear escalation advice	User feedback, questionnaires to understand in-patient feedback, subsequent high-risk follow-up	5	140	Revisit consent screen in plain language with explicit opt-in for data use in ML; add in-trip privacy dashboard or note showing what data is stored and why; implement data deletion on user request	Ruid	February						0	
###													0								0	
###													0								0	

RPN Column contains a formula to auto-calculate:
=IF(OR(A11="","",B11="","",C11="")),0, (B11*A11*C11)

DO5 – Privacy, Consent, and Access Control Documentation

To verify DO5, the consent workflow must be inspected, and role-based access must be tested to ensure there are 0 unauthorized access paths during role testing and 100% of the personal data access are gated behind explicit user consent. The following industry standards will be referenced to:

- Google Play User Data Policy [26]
- PIPEDA Principles [27]

DO6 – Risk Management File Output Verification Plan

FMEA identified in Chapter 2 needs to be verified through fault injection, stress and edge-case testing. All the identified risks and failure modes must be mapped to mitigation strategies. Stress and edge-case testing will be used to verify the mitigation effectiveness to meet $\geq 95\%$ mitigation effectiveness under simulated fault conditions. The following documents must be aligned with to meet the industrial standard:

- ISO 14971 – Risk Management [28]
- Google App Stability Metrics [29]

Barriers Delaying Dental Care

COST

Dental care is one of the **most unaffordable** health services [4]

DENTAL FEAR

Anxiety/fear cause many individuals to **avoid or postpone** treatment[5]

AWARENESS

Lack of awareness of early symptoms & preventive care contributes to **delayed treatment** [5].



SOCIONOMIC INEQUALITIES

Large disparities in healthcare systems, income & education levels [5]

ACCESS

Limited resources and access in rural/remote areas [5]

LIMITED EARLY SELF-SCREENING TOOLS

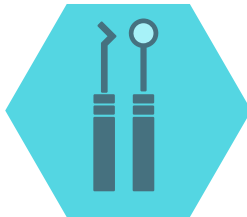
Lack of simple tools for early self-screening prevents timely recognition of problems.

Why Oral Health Matters

3.7 Billion people affected [1]



1+ billion periodontal (gum & bone inflammations) disease cases [2]



Adversely affect an individual's emotional state & social interactions [3]



2.37 Billion untreated caries
Most common global condition [1]



Major effects on speaking, chewing, swallowing and confidence in smiling [3]



Most Affected Populations



Groups

- Lower-income & Middle-income
- Uninsured
- Minority
- Immigrants



CHILDREN

Children from low-income families:
2x higher untreated cavity rates



ELDERS

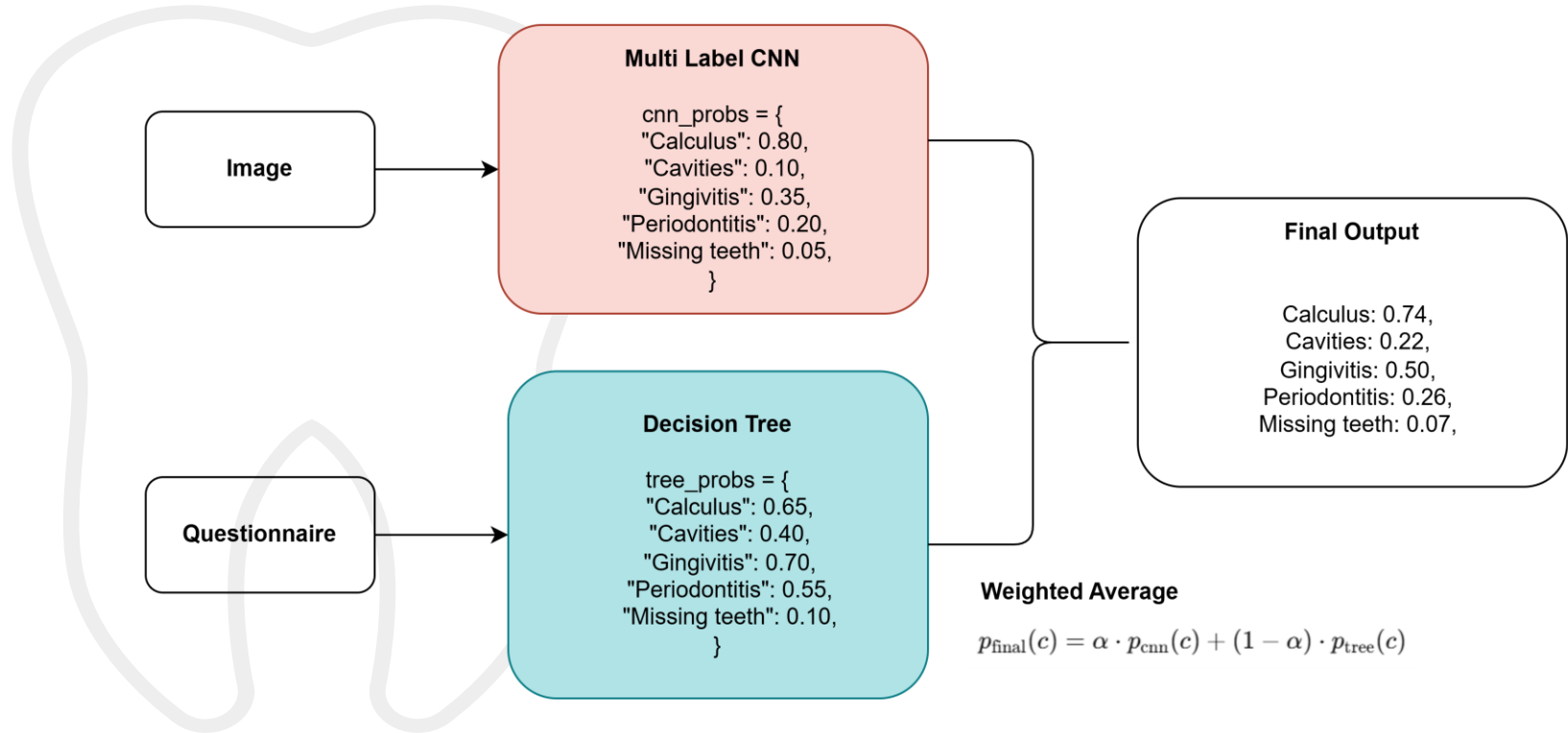
Older adults face literacy and resource challenges



Rural Populations

Rural populations lack consistent access to dentists

Proposed Solution



Prototype

Binary Classification (Healthy vs Nonhealthy)

Problem Definition: Build a prototype capable of detecting Dental Calculus from intraoral tooth images.

Dataset:

Total images (307):

- Calculus: 159
- Healthy: 148

Dataset split:

- 80% training (245 images)
- 20% testing (62 images)



Preprocessing & Augmentation

- Images resized to 224×224
- Normalized using ImageNet mean and standard deviation
- Augmentation applied to increase robustness:
 - Random rotation ($\pm 15^\circ$)
 - Brightness jitter (0.9 – 1.1)
 - Horizontal flipping

Training Details

- **Loss:** Cross-Entropy Loss
- **Optimizer:** Adam
- Train for 25 epochs
- Track training/test accuracy & loss each epoch
- Save best-performing model based on test accuracy



MobileNetV3-Small

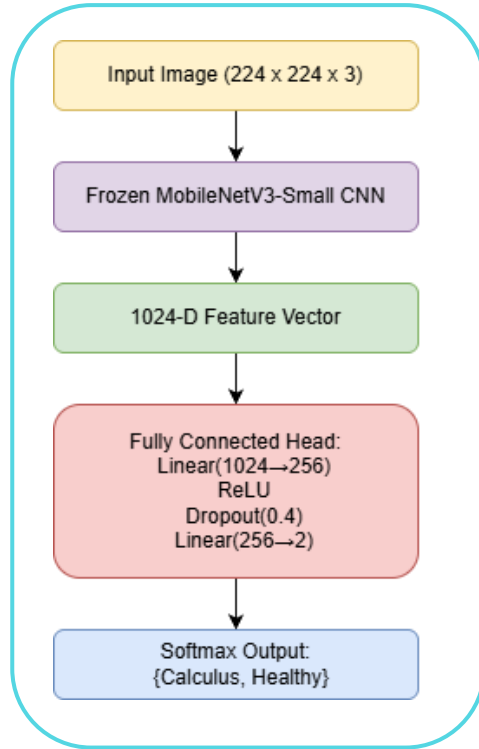
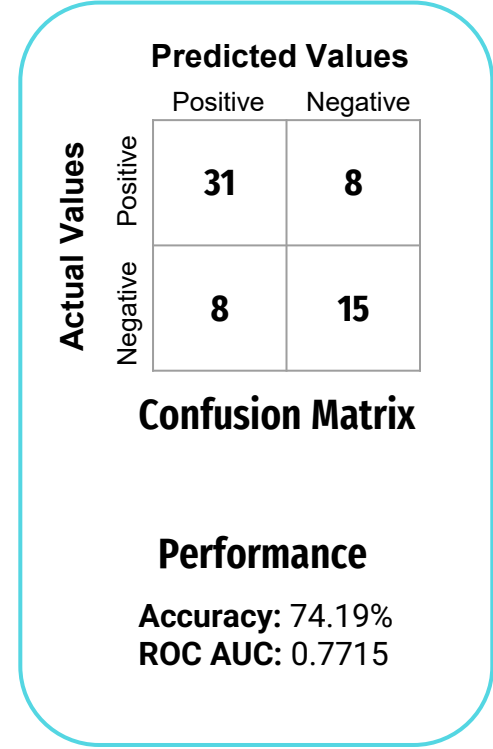


Figure 4. MobileNetV3-Small ML Architecture



Figure 5. MobileNetV3-Small Accuracy and Loss graphs



ResNet18

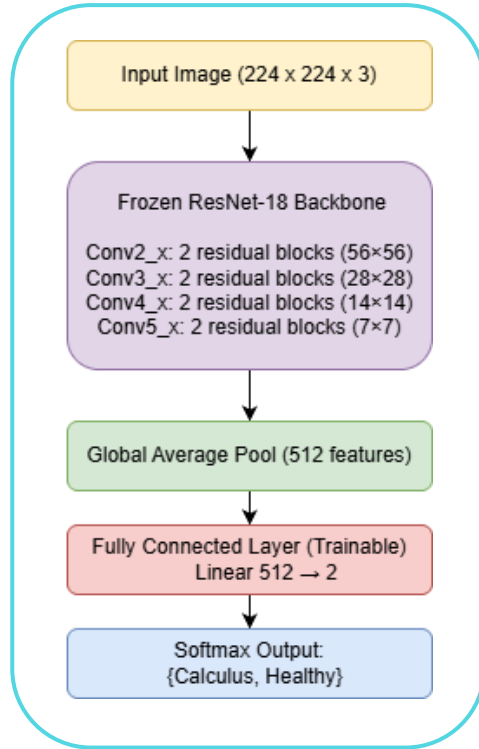


Figure 6. ResNet-18
ML Architecture

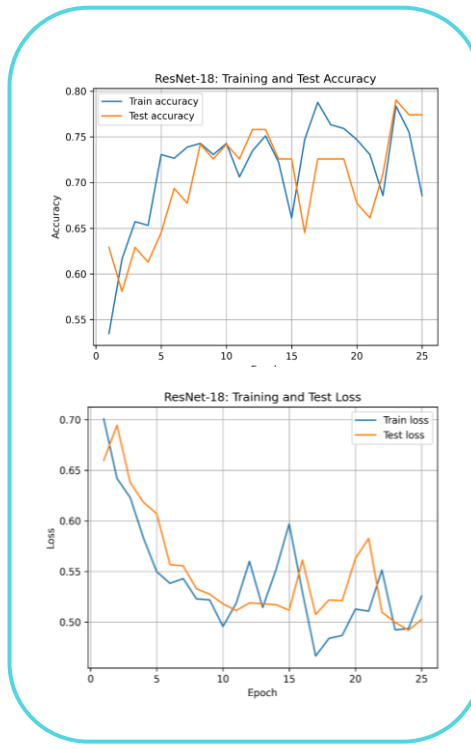


Figure 7. ResNet-18
Accuracy and Loss graphs

		Predicted Values	
		Positive	Negative
Actual Values	Positive	22	8
	Negative	5	27

Confusion Matrix

Performance

Accuracy: 79.03%
ROC AUC: 0.8208

Testing



Healthy



Light Dental Calculus



Heavy Dental Calculus

Results

Gradient-Weighted Class Activation Mapping (Grad-CAM)

**MobileNetV3
-Small**

Predicted: Healthy



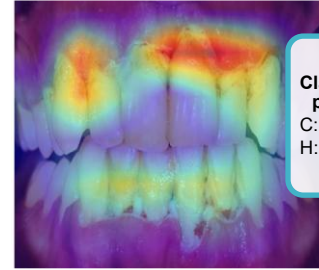
Class probabilities:
C: 0.1604
H: 0.8396

Predicted: Calculus



Class probabilities:
C: 0.7974
H: 0.2026

Predicted: Calculus



Class probabilities:
C: 0.9957
H: 0.0043

ResNet-18

Predicted: Healthy



Class probabilities:
C: 0.1793
H: 0.8207

Predicted: Calculus



Class probabilities:
C: 0.8297
H: 0.1703

Predicted: Calculus



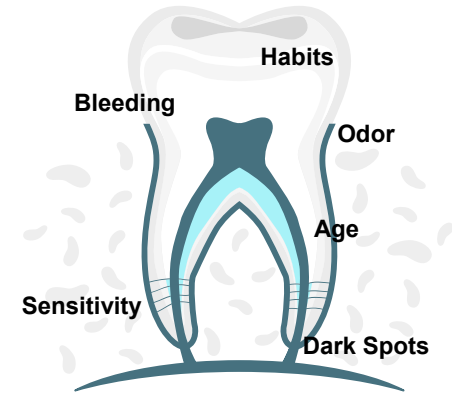
Class probabilities:
C: 0.9618
H: 0.0382

C= Calculus; H = Healthy

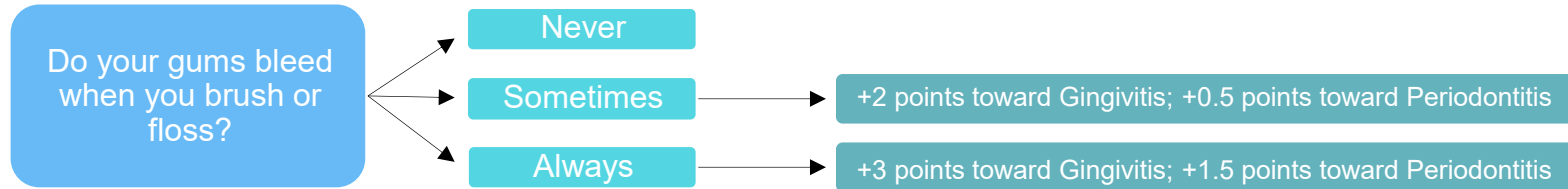
Questionnaire based Decision Tree

How it works:

1. User answers **10 clinically-relevant questions** about symptoms and history
2. Each answer contributes weighted “risk points” toward:
 - **Calculus**
 - **Cavities**
 - **Gingivitis**
 - **Periodontitis**
 - **Missing teeth**
3. The system converts scores into probabilities
4. **Output:** Estimated risk levels for each condition



Example:



MobileNet V3-Small

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